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الجامعة العربية المفتوحة
Arab Open University
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MT132
Linear Algebra

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 16/April/2026

Number of Exam Pages: 3 pages (including this cover sheet)	**Time Allowed:** 90 Minutes

Instructions:

- Total Marks: **30**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Q–1: [4×2 marks] Choose the correct answer:

1.1: If a matrix A of the size 3×3 is invertible and $\det(A) = 2$. What is $|-2A^{-1}|$?

a) 4

b) -4

c) $\frac{1}{4}$

d) $-\frac{1}{4}$

e) None of the above

$-2^3\left(\frac{1}{2}\right)$

1.2: If X_1 and X_2 are solutions for the linear system $AX = 0$, then $sX_1 - tX_2$ is also a solution if

a) $s - t = 1$

b) $s + t = 1$

c) $s - t = 0$

d) s and t are any real number

e) None of the above

1.3: The inverse of the identity matrix of the size 2×2 is

a) $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

c) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

d) $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

e) None of the above

1.4: The linear system $AX = 0$ has a nontrivial solution if

a) A is nonsingular

b) A is invertible

c) $A = \begin{bmatrix} -1 & 3 \\ 2 & -6 \end{bmatrix}$

d) $|A| = -1$

e) None of the above

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Q–2: [8 marks] Answer the following questions as True or False (1 mark), Justifying your answer (3 marks):

2.1: The product of two skew-symmetric matrices is always skew-symmetric matrix.

2.2: If $|A| = |B|$, then $A = B$.

Each question is worth 7 marks. Answer the following questions:

Q–3 [4+2+1 marks]:

Consider the linear system: $\begin{cases} -y + z = -2 \\ x - y = 3 \\ -x + 3y - z = 0 \end{cases}$,

a) Find the inverse of the coefficient matrix of the linear system, A.

b) Solve the linear system.

c) Is the linear system independent? Justify.

Q–4 [4+2+1 marks]:

Let $A = \begin{bmatrix} 3 & 0 & 0 & -4 \\ 0 & 1 & -1 & 0 \\ -2 & 0 & 0 & 3 \\ 0 & 4 & 2 & -1 \end{bmatrix}$.

a) Find $|A^{-1}|$.

b) Find $|3A^{-1}A^T|$.

c) Is the matrix A singular? Justify.

End of questions

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